SignKG: A Spatial-Aware Framework for Constructing Urban Business Knowledge Graphs from Vietnamese Street Signboards

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Abstract. Street-level signboards contain valuable urban business information, yet conventional OCR-based systems typically return isolated text strings without identifying their semantic roles or relationships. This limitation is particularly challenging for Vietnamese signboards because of decorative fonts, complex diacritics, irregular layouts, and ambiguous text content. This paper proposes *SignKG*, a spatial-aware framework that transforms Vietnamese street-signboard images into structured urban business knowledge graphs. SignKG integrates visual perception, spatial-aware entity extraction, constrained relation inference, and graph optimization. The framework jointly exploits lexical semantics, textbox positions, neighborhood context, business rules, and symbolic compatibility constraints to identify business entities and construct semantically consistent relations. Experiments on a real-world Vietnamese signboard dataset show that SignKG achieves entity-, relation-, and graph-level F_1 scores of 0.910, 0.874, and 0.851, respectively. It also obtains a valid-edge ratio of 0.967 and a store-connectivity ratio of 0.951, outperforming rule-based, text-only, layout-aware, and LLM-based baselines. Ablation results confirm that spatial information is critical for entity disambiguation, while symbolic constraints improve graph validity and structural consistency.

Keywords: Urban Business Knowledge Graphs · Vietnamese Street Signboards · Spatial-Aware Information Extraction · Knowledge Graph Construction · Optical Character Recognition · Neuro-Symbolic Reasoning · Urban Computing

1 Introduction

Street signboards are among the most visible sources of business information in urban environments. They communicate to human observers the existence of stores, services, brands, addresses, and contact information. As such, they constitute a rich and continuously evolving source of urban knowledge. However, this knowledge is mostly embedded in unstructured visual content, making it difficult to exploit for digital maps, business search, recommendation systems, and smart-city analytics.

Recent advances in computer vision and optical character recognition have significantly improved the ability to detect signboards and recognize text from street imagery. Given a signboard image, modern OCR systems can often recover a set of textual fragments with reasonable accuracy. However, recognizing text is fundamentally different

from understanding business knowledge. For instance, consider a Vietnamese street signboard from which an OCR system extracts

$$T = \{\texttt{PHO HA NOI}, \texttt{PHO BO}, \texttt{123 NGUYEN VAN CU}, \texttt{0909XXXXXX}\}.$$

Although the textual content is recovered, the system still does not know whether `PHO HA NOI` denotes a brand, a service, or a location name. Likewise, it does not know whether `PHO BO` is a service description or a business category, nor how the phone number and address are related to the corresponding store. Thus, OCR provides symbols, but not structured knowledge.

This limitation becomes critical in urban-scale applications. A digital map or a business intelligence system cannot rely on isolated text strings alone. Instead, it requires structured entities and relations, such as

$$\texttt{Store} \xrightarrow{\texttt{HAS_BRAND}} \texttt{Brand}, \qquad \texttt{Store} \xrightarrow{\texttt{PROVIDES}} \texttt{Service},$$

$$\texttt{Store} \xrightarrow{\texttt{LOCATED_AT}} \texttt{Address}, \qquad \texttt{Store} \xrightarrow{\texttt{HAS_GEO}} \texttt{Location}.$$

In other words, the target representation is not a list of recognized words, but a business knowledge graph.

Formally, let I denote a signboard image. Most existing signboard understanding approaches focus on learning a mapping $f : I \to T$, where $T = \{t_1, t_2, \ldots, t_n\}$ is the set of recognized text segments. While useful, this representation remains incomplete because it ignores entity types, semantic dependencies, and spatial relations. In contrast, urban business knowledge acquisition requires a mapping $F : I \to G$, where $G = (V, E)$ is a business knowledge graph. Here, V denotes a set of business entities, including stores, brands, services, categories, addresses, phone numbers, and geographic locations, while E denotes semantic and spatial relations among these entities. *Constructing such a graph from signboard imagery is challenging for three reasons.* First, Vietnamese street signboards are highly diverse in terms of fonts, colors, layouts, and illumination conditions. Second, the output of OCR is often noisy and fragmented, especially for Vietnamese text with diacritics. Third, and most importantly, the semantic role of a text fragment is not determined by its lexical content alone. Its position on the signboard also matters. For example, a large text segment located near the visual center is often the business brand, whereas smaller text located near the bottom is more likely to be an address, a phone number, or a service description. Therefore, signboard understanding should be treated as a joint semantic–spatial reasoning problem rather than a purely textual extraction task.

To address this problem, we propose SignKG, a spatial-aware framework for constructing urban business knowledge graphs from Vietnamese street signboards. The central idea is to exploit not only the recognized text but also its spatial layout and geographic context. Each extracted text segment t_i is associated with a bounding region $b_i = (x_i, y_i, w_i, h_i)$, and the pair (t_i, b_i) is used to infer the entity type and its relation to other entities. Thus, instead of treating OCR outputs as independent strings, SignKG interprets them as spatially grounded semantic units.

The proposed framework transforms an input signboard image into a graph-structured representation through three main stages. First, visual processing is used to localize

the signboard, correct perspective distortion, and recognize Vietnamese text. Second, spatial-aware information extraction identifies business entities by jointly considering textual semantics and layout information. Third, the extracted entities are linked into a business knowledge graph that supports semantic search, spatial discovery, and downstream urban analytics.

The contributions of this paper are summarized as follows:

- We formulate the construction of urban business knowledge graphs from street signboards as a structured knowledge acquisition problem.
- We introduce a spatial-aware extraction mechanism that exploits both textual semantics and signboard layout for identifying business entities.
- We construct a business knowledge graph linking stores, brands, services, categories, addresses, phone numbers, and geographic locations.
- We evaluate the proposed framework on real-world Vietnamese street signboards and demonstrate its potential for urban business mapping and smart-city applications.

2 Problem Formulation

The objective of SignKG is to transform an unstructured street signboard image into a structured urban business knowledge graph. This objective is different from conventional signboard OCR. An OCR system only returns a set of recognized strings, whereas a knowledge graph construction system must identify what these strings mean and how they are related. For example, from a Vietnamese signboard, OCR may return $T = \{\texttt{PHO HA NOI}, \texttt{PHO BO}, \texttt{123 NGUYEN VAN CU}, \texttt{0909XXXXXX}\}$. Although these strings are useful, they are not yet knowledge. The system still has to determine whether `PHO HA NOI` is a brand, whether `PHO BO` is a service, whether `123 NGUYEN VAN CU` is an address, and how these entities are connected to the same store. Therefore, the problem studied in this paper is a structured knowledge acquisition problem from visual data.

2.1 Spatial Text Segments

Let I be a street signboard image. After detection, rectification, and OCR, the image is represented as a set of spatial text segments.

Definition 1 (Spatial Text Segment). *A spatial text segment is a tuple $s_i = (t_i, b_i)$, where t_i is the recognized textual content and $b_i = (x_i, y_i, w_i, h_i)$ is the bounding box describing the spatial position and size of the text segment in the signboard image.*

The set of all extracted spatial text segments from image I is denoted by $S(I) = \{s_1, s_2, \ldots, s_n\}$. This representation is richer than a standard OCR output because it preserves not only what is written, but also where it is written. This distinction is crucial for signboard understanding. In real signboards, large central text often denotes the business brand, while smaller text located near the lower region usually corresponds to an address, a phone number, or a service description. Hence, the spatial layout of text provides useful evidence for semantic interpretation.

2.2 Business Entities

The goal of SignKG is to infer business entities from spatial text segments. A business entity is not simply a text string, but a typed object that can participate in graph relations.

Definition 2 (Business Entity). *A business entity is a pair $e = (v, \tau)$, where v denotes the entity value and $\tau \in \mathcal{T}$ denotes its semantic type.*

In this work, the entity type vocabulary is defined as $\mathcal{T} = \{Store, Brand, Service, Category, Address, Phone, Location\}$ For example, (`Pho Ha Noi`, $Brand$), (`Pho Bo`, $Service$), and (`123 Le Duan`, $Address$) are different business entities, even though they may all appear as plain strings in the OCR output. Given a signboard image I, the entity extraction task is to infer a set of typed entities $V(I) = \{e_1, e_2, \ldots, e_m\}$ from the spatial text segment set $S(I)$.

2.3 Spatial-Aware Entity Assignment

A central assumption of SignKG is that the semantic type of a text segment depends on both textual content and spatial layout. This assumption reflects how humans read signboards. A person does not interpret a signboard by reading isolated strings only; they also use visual cues such as position, font size, alignment, and relative ordering. For this reason, SignKG models entity assignment as a spatial-aware inference task.

Definition 3 (Spatial-Aware Entity Assignment). *Given a spatial text segment $s_i = (t_i, b_i)$, the spatial-aware entity assignment problem consists of finding a semantic type $\tau_i \in \mathcal{T}$ for t_i by considering both its textual content t_i and its bounding box b_i.*

Formally, the optimal semantic type is defined as $\tau_i^* = \arg\max_{\tau \in \mathcal{T}} P(\tau \mid t_i, b_i)$. This formulation differs from conventional text classification, which estimates $P(\tau \mid t_i)$ and ignores the visual layout. The use of b_i is important because the same textual fragment may play different roles under different spatial contexts. For example, the phrase `HA NOI` may be part of a brand in `PHO HA NOI`, but it may also indicate a location in another signboard. Similarly, a number string can be a phone number, a house number, or part of a service name depending on where it appears and how it is arranged with nearby text. Therefore, spatial information acts as an additional semantic signal for reducing ambiguity.

2.4 Urban Business Knowledge Graph

After business entities are identified, they must be organized into a structured representation that supports querying and reasoning. A knowledge graph is suitable for this purpose because it explicitly represents entities as nodes and relations as labeled edges.

Definition 4 (Urban Business Knowledge Graph). *An urban business knowledge graph is a directed labeled graph $G = (V, R, E)$, where V is a set of business entities, R is a set of relation types, and $E \subseteq V \times R \times V$ is a set of semantic relations between entities.*

The relation vocabulary used in this work is defined as $R = \{HAS_BRAND, PROVIDES, BELONGS_TO, LOCATED_AT, HAS_PHONE, HAS_GEO\}$. For example, a store can be connected to its brand by $(Store, HAS_BRAND, Brand)$, to its services by $(Store, PROVIDES, Service)$, to its category by $(Store, BELONGS_TO, Category)$, to its address by $(Store, LOCATED_AT, Address)$, to its phone number by $(Store, HAS_PHONE, Phone)$, and to its geographic coordinate by $(Store, HAS_GEO, Location)$. Compared with a list of OCR strings, this graph representation is more informative because it describes not only the extracted entities but also their business meaning and interconnections.

2.5 SignKG Construction Problem

We now define the main problem addressed in this paper. Let $\mathcal{I}$ denote the space of street signboard images and let $\mathcal{G}$ denote the space of urban business knowledge graphs.

Definition 5 (SignKG Construction). *Given a signboard image $I \in \mathcal{I}$, the SignKG construction problem consists of generating a graph $G \in \mathcal{G}$ that best represents the business knowledge contained in I.*

Formally, SignKG seeks the optimal graph $G^* = \arg\max_{G \in \mathcal{G}} P(G \mid I)$. Since a knowledge graph is determined by its entities and relations, this probability can be decomposed as $P(G \mid I) = P(V \mid I)P(E \mid V, I)$. The first term $P(V \mid I)$ corresponds to business entity extraction, while the second term $P(E \mid V, I)$ corresponds to relation construction. Thus, SignKG jointly addresses two coupled tasks: identifying typed business entities from signboard imagery and linking these entities into a semantically meaningful graph.

Problem 1 (Urban Business Knowledge Acquisition from Signboards). Given a signboard image I, a set of spatial text segments $S(I) = \{(t_i, b_i)\}_{i=1}^{n}$, an entity vocabulary $\mathcal{T}$, and a relation vocabulary R, find a knowledge graph $G^* = (V^*, R, E^*)$ such that $G^* = \arg\max_{G \in \mathcal{G}} P(G \mid I)$.

This formulation clarifies the distinction between SignKG and standard OCR-based pipelines. A standard pipeline learns a mapping $f : I \to T$, where T is a set of recognized text strings. In contrast, SignKG learns a mapping $F : I \to G$, where G is a graph-structured representation of business knowledge. The output of SignKG is therefore not only readable by humans, but also computable by downstream systems for semantic search, spatial querying, business discovery, and urban analytics.

3 Spatial-Aware Knowledge Modeling

The previous section formulated SignKG as the problem of finding a graph G^* that maximizes $P(G \mid I)$. This section specifies how spatial information is modeled and used as evidence for constructing such a graph. The main idea is simple but important: in street signboards, text is not only a sequence of words, but also a spatial object. Its meaning depends on where it appears, how large it is, and how it is arranged with respect to other text segments.

Let $S(I) = \{s_1, s_2, \ldots, s_n\}$ be the set of spatial text segments extracted from a signboard image I, where each segment is defined as $s_i = (t_i, b_i)$ and $b_i = (x_i, y_i, w_i, h_i)$. A conventional text-based system estimates the semantic type of s_i only from t_i. In contrast, SignKG estimates the semantic type from both t_i and b_i. This distinction is crucial because a signboard is visually organized to guide human interpretation. Large central text is often designed to attract attention and usually denotes the brand, while smaller text near the bottom is more likely to describe addresses, phone numbers, or services. Therefore, spatial layout provides useful prior knowledge for entity inference.

Definition 6 (Normalized Spatial Descriptor). *Given a spatial text segment $s_i = (t_i, b_i)$ with $b_i = (x_i, y_i, w_i, h_i)$ and a signboard image of width W and height H, the normalized spatial descriptor of s_i is defined as $\rho_i = (\bar{x}_i, \bar{y}_i, \bar{w}_i, \bar{h}_i, \bar{a}_i)$, where $\bar{x}_i = x_i/W$, $\bar{y}_i = y_i/H$, $\bar{w}_i = w_i/W$, $\bar{h}_i = h_i/H$, and $\bar{a}_i = (w_i h_i)/(WH)$.*

The descriptor ρ_i makes the spatial representation independent of image resolution. This is necessary because signboard images may be captured from different devices, distances, and viewpoints. The value $\bar{a}_i$ measures the relative visual dominance of the segment. In practice, a segment with a large $\bar{a}_i$ and a central position is more likely to be a brand, whereas a segment with a small $\bar{a}_i$ and a lower position is more likely to be an address or phone number.

Definition 7 (Spatial Neighborhood). *Given two spatial text segments $s_i = (t_i, b_i)$ and $s_j = (t_j, b_j)$, let $d(b_i, b_j)$ denote the Euclidean distance between the centers of b_i and b_j. For a distance threshold $\delta > 0$, the spatial neighborhood of s_i is defined as $\mathcal{N}_\delta(s_i) = \{s_j \in S(I) \mid j \neq i \text{ and } d(b_i, b_j) \leq \delta\}$.*

The spatial neighborhood captures local layout context. This is useful because the role of a text segment is often determined by nearby segments. For example, a number string located next to the word `Hotline` is likely to be a phone number, while a number string located before a street name is likely to be part of an address. Similarly, a service phrase located close to a brand may describe what the store provides. Thus, SignKG does not interpret text segments independently, but uses their local spatial context.

Definition 8 (Spatial Context). *The spatial context of a segment s_i is defined as $c_i = (t_i, \rho_i, \mathcal{N}_\delta(s_i))$, where t_i is the textual content, ρ_i is its normalized spatial descriptor, and $\mathcal{N}_\delta(s_i)$ is its spatial neighborhood.*

Proposition 1 (Spatial Context Reduces Semantic Uncertainty). *Let τ be the semantic type of a text segment, t its textual content, and $c = (t, \rho, \mathcal{N}_\delta)$ its spatial context. Then $H(\tau|c) \leq H(\tau|t)$.*

Proof. Since $c = (t, \rho, \mathcal{N}_\delta)$ contains t together with additional spatial evidence, we have $H(\tau|c) = H(\tau|t, \rho, \mathcal{N}_\delta)$. By the monotonicity of conditional entropy, conditioning on additional variables cannot increase uncertainty. Equivalently, $H(\tau|t) - H(\tau|t, \rho, \mathcal{N}_\delta) = I(\tau; \rho, \mathcal{N}_\delta|t) \geq 0$, where $I(\cdot;\cdot|\cdot)$ denotes conditional mutual information. Hence $H(\tau|c) \leq H(\tau|t)$.

Based on this context, spatial-aware entity inference is formulated as $\tau_i^* = \arg\max_{\tau \in \mathcal{T}} P(\tau \mid c_i)$. Since c_i contains both textual and spatial evidence, this formulation is more expressive than $\arg\max_{\tau \in \mathcal{T}} P(\tau \mid t_i)$. It allows SignKG to distinguish cases where lexical content alone is ambiguous. For instance, the phrase `HA NOI` may be interpreted as part of a brand when it appears in a large central text block, but as a location-related phrase when it appears in an address-like line.

After entity types are inferred, SignKG must determine which entities should be connected. Relation construction is also spatially grounded. Entities appearing on the same signboard and sharing consistent spatial organization are more likely to be related to the same store. For example, a brand entity and a phone entity extracted from the same signboard should be connected through the store node, while a service entity should be linked by the relation $PROVIDES$.

Definition 9 (Relation Compatibility). *Let $e_i = (v_i, \tau_i)$ and $e_j = (v_j, \tau_j)$ be two business entities. A relation $r \in R$ is compatible with (e_i, e_j) if the type pair (τ_i, τ_j) satisfies the domain and range constraints of r.*

For example, HAS_BRAND is compatible with $(Store, Brand)$, $PROVIDES$ is compatible with $(Store, Service)$, $LOCATED_AT$ is compatible with $(Store, Address)$, HAS_PHONE is compatible with $(Store, Phone)$, and HAS_GEO is compatible with $(Store, Location)$. These constraints prevent invalid graph edges, such as connecting a phone number to a category through $LOCATED_AT$. Therefore, relation compatibility acts as a symbolic constraint that improves the consistency of the generated knowledge graph.

Definition 10 (Spatial-Aware Relation Assignment). *Given two entities e_i and e_j extracted from image I, the relation assignment problem consists of finding $r^* = \arg\max_{r \in R} P(r \mid e_i, e_j, I)$ subject to the compatibility constraint that r is valid for the type pair (τ_i, τ_j).*

Proposition 2 (Type-Sound Relation Filtering). *Let $\hat{E} \subseteq V \times R \times V$ be a candidate relation set and let $E = \{(e_i, r, e_j) \in \hat{E} \mid \Gamma(\tau_i, \tau_j, r) = 1\}$. Then every relation in E is type-compatible.*

Proof. By construction, a triple (e_i, r, e_j) is inserted into E only if $\Gamma(\tau_i, \tau_j, r) = 1$. According to the definition of Γ, this condition holds exactly when relation r satisfies the domain and range constraints of the type pair (τ_i, τ_j). Therefore, every triple in E is type-compatible.

This definition shows that relation construction is not treated as arbitrary edge generation. It is constrained by entity types and supported by the visual source from which the entities are extracted. In this work, the store entity acts as the central anchor of the graph. Other entities, including brand, service, category, address, phone number, and geographic location, are linked to this anchor through predefined relation types. This design reflects the natural structure of a business signboard: most information on the signboard describes one business object.

The resulting graph must satisfy basic consistency requirements. First, every non-store entity extracted from a signboard should be reachable from a store node. Second,

every edge must satisfy relation compatibility. Third, geographic information, if available from image metadata, should be attached to the store entity rather than to arbitrary text nodes. These requirements ensure that the generated graph is not only a set of extracted triples, but a coherent business representation.

Definition 11 (Graph Consistency). *A generated graph $G = (V, R, E)$ is consistent if every edge $(e_i, r, e_j) \in E$ satisfies the compatibility constraints of r, and every extracted business entity is either a store entity or connected to a store entity through at least one valid relation.*

Proposition 3 (Store-Anchored Graph Connectivity). *Assume that a signboard contains one dominant store entity e_s and that every non-store entity $e_i \in V \setminus \{e_s\}$ is linked to e_s by at least one compatible relation. Then the generated graph $G = (V, R, E)$ is weakly connected.*

Proof. For every entity $e_i \in V \setminus \{e_s\}$, there exists at least one edge connecting e_i and e_s. If edge directions are ignored, every node has a path of length one to e_s. Therefore, for any two nodes $e_i, e_j \in V$, there exists a path $e_i \rightarrow e_s \rightarrow e_j$ in the underlying undirected graph. Hence G is weakly connected.

Proposition 4 (Consistency of SIGNKG Graph Construction). *Let $G = (V, R, E)$ be generated by Algorithm 1. If every inserted edge satisfies $\Gamma(\tau_i, \tau_j, r) = 1$ and every non-store entity is connected to a store entity, then G is consistent.*

By Algorithm 1, an edge (e_i, r, e_j) is inserted into E only when $\Gamma(\tau_i, \tau_j, r) = 1$. Under this model, SignKG constructs a graph by maximizing the joint plausibility of entity assignments and relation assignments while satisfying graph consistency. The structured objective can be written as $G^* = \arg\max_{G \in \mathcal{G}} P(V \mid I)P(E \mid V, I)$, subject to the constraint that G is consistent. This formulation connects the visual input, spatial layout, semantic interpretation, and graph construction into a single knowledge modeling view.

In summary, the spatial-aware knowledge model provides the formal basis for SignKG. It explains why OCR strings alone are insufficient, why bounding boxes and layout should be treated as semantic evidence, and why graph construction must obey relation compatibility constraints. The next section describes how this model is implemented by the proposed SignKG framework.

4 The SignKG Framework

4.1 Framework Overview

The objective of SIGNKG is to solve the urban business knowledge acquisition problem by transforming a street signboard image I into an urban business knowledge graph G. Instead of directly estimating $P(G|I)$, which is highly intractable due to the heterogeneous nature of visual and semantic information, the proposed framework decomposes the inference process into a sequence of four hierarchical stages. Each stage estimates

an intermediate representation with increasing semantic abstraction, thereby progressively reducing the uncertainty of the final graph construction.

Given an input image I, SignKG first estimates a spatial text representation $S(I) = \{(t_i, b_i)\}_{i=1}^{n}$ through visual perception, where t_i and b_i denote the recognized text and its corresponding spatial location, respectively. Based on $S(I)$, the framework infers a set of typed business entities $V(I)$ through spatial-aware semantic inference, establishes semantic relations $E(I)$ through spatial-aware relation inference, and finally constructs an urban business knowledge graph $G = (V, R, E)$. Accordingly, the overall inference process is formulated as

$$P(G|I) = P(G|E)P(E|V)P(V|S)P(S|I), \tag{1}$$

where $P(S|I)$ models visual perception, $P(V|S)$ models spatial-aware semantic inference, $P(E|V)$ models relation inference, and $P(G|E)$ models knowledge graph construction. This probabilistic decomposition provides a unified interpretation of the proposed framework by assigning a well-defined inference objective to each stage. In contrast to conventional OCR-based pipelines that learn a mapping $f : I \to T$, where T denotes recognized text strings, SignKG learns a mapping $F : I \to G$, where G is a structured knowledge graph. Consequently, OCR is regarded as an intermediate representation rather than the final objective. The framework progressively transforms raw visual observations into structured business knowledge through the inference chain

$$I \to S(I) \to V(I) \to E(I) \to G, \tag{2}$$

which preserves visual evidence, semantic interpretation, and spatial consistency throughout the entire knowledge acquisition process. Figure 1 illustrates the overall architecture, while the following subsections describe each inference stage in detail. Figure 1 illustrates the overall architecture of SIGNKG. The framework starts from an input street signboard image and progressively produces $S(I)$, $V(I)$, $E(I)$, and G. The following subsections describe the four stages in detail.

4.2 Visual Perception

The objective of the visual perception stage is to infer a spatially consistent textual representation from an input street signboard image. Rather than directly recognizing isolated text strings, this stage estimates a structured observation that preserves both textual content and geometric layout. Such a representation serves as the visual evidence for all subsequent semantic inference tasks.

Let $I \in \mathcal{I}$ denote an input street signboard image and let $S(I) = \{(t_i, b_i)\}_{i=1}^{n}$ denote the corresponding set of spatial text segments, where t_i is the recognized text and $b_i = (x_i, y_i, w_i, h_i)$ is its bounding box. The visual perception problem is formulated as estimating the most probable spatial representation $S^* = \arg\max_S P(S|I)$, where $P(S|I)$ measures the likelihood that the extracted spatial text representation faithfully describes the visual content of image I.

Unlike conventional OCR pipelines, the objective of SignKG is not merely to maximize text recognition accuracy. Instead, the extracted representation should satisfy three properties: (i) textual completeness, ensuring that business information is not omitted;

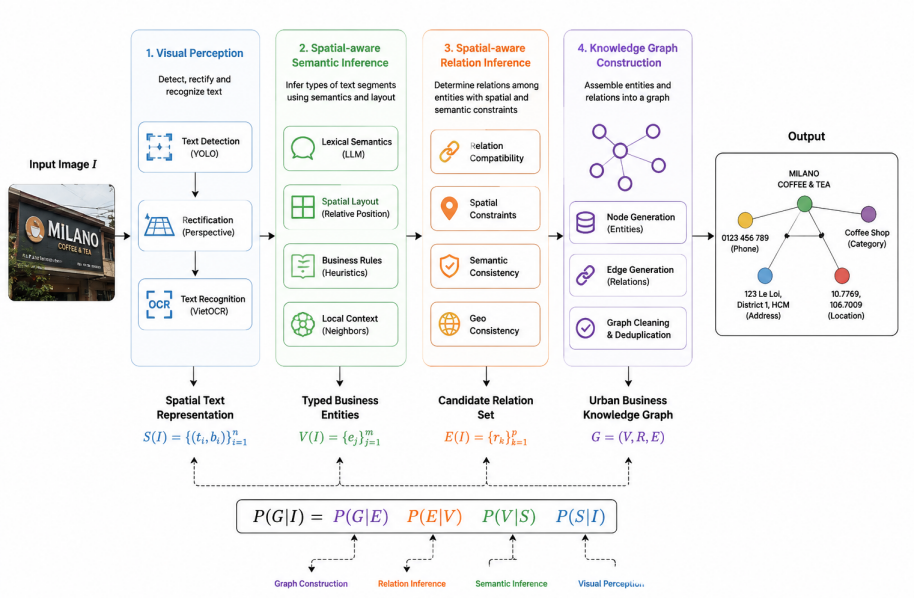


Fig. 1: Overview of the proposed SIGNKG framework. The overall objective $P(G|I)$ is decomposed into four successive inference stages: visual perception $P(S|I)$, spatial-aware semantic inference $P(V|S)$, spatial-aware relation inference $P(E|V)$, and knowledge graph construction $P(G|E)$. This hierarchical decomposition progressively transforms an input street signboard image into a structured urban business knowledge graph while preserving both semantic meaning and spatial organization.

(ii) geometric consistency, preserving the original spatial arrangement of text segments; and (iii) semantic readiness, providing sufficient information for downstream entity and relation inference. Consequently, the output of the visual perception stage is a structured observation rather than a collection of independent text strings. To estimate $P(S|I)$, the framework decomposes visual perception into three successive inference steps. First, the signboard region is localized within the input image by estimating a set of candidate regions $\mathcal{B} = \{b_1, \ldots, b_k\}$. This localization step suppresses irrelevant background information and restricts subsequent processing to visually informative regions. Second, the detected signboard is geometrically normalized through perspective rectification. Let $\Psi(\cdot)$ denote the rectification operator. The normalized signboard image is obtained as $\hat{I} = \Psi(I)$, where geometric distortions caused by viewpoint variation are minimized. Third, optical character recognition estimates the structured observation $S(\hat{I})$ by jointly recovering textual content and its spatial locations. Accordingly, the overall visual perception process can be expressed as $I \rightarrow \mathcal{B} \rightarrow \hat{I} \rightarrow S(I)$, which progressively transforms raw visual observations into a structured spatial representation suitable for semantic reasoning. An important characteristic of this formulation is that the output remains independent of the underlying perception model. Regardless of whether the detector or OCR engine is replaced by a stronger alternative, the subsequent

semantic inference stage always receives the same structured representation $S(I)$. This abstraction clearly separates perception from reasoning and substantially improves the modularity of the proposed framework.

In the current implementation, signboard localization is performed using YOLOv8, geometric normalization is achieved through perspective rectification, and Vietnamese text recognition is carried out using VietOCR. These components provide one practical realization of the inference process, while the mathematical formulation of SignKG remains independent of any particular visual backbone.

4.3 Spatial-aware Semantic Inference

The objective of the second stage is to transform the structured visual observations obtained from the visual perception stage into typed business entities. Unlike conventional information extraction, where each text segment is interpreted independently, SignKG formulates semantic extraction as a spatial-aware inference problem in which textual semantics and visual layout are jointly considered. The underlying assumption is that the semantic role of a text segment cannot always be determined from its lexical content alone. Instead, its surrounding context and spatial organization provide complementary evidence that substantially reduces semantic ambiguity.

Let $S(I) = \{(t_i, b_i)\}_{i=1}^{n}$ denote the spatial text representation extracted from image I, where t_i is the recognized text and b_i is its spatial descriptor. The objective of this stage is to infer the corresponding business entity set $V(I) = \{e_j\}_{j=1}^{m}$, where each entity is represented by $e = (v, \tau)$ with $\tau \in \mathcal{T}$. Formally, the semantic inference problem is defined as $V^* = \arg\max_V P(V|S)$, where $P(V|S)$ denotes the posterior probability that the inferred entity set correctly explains the observed spatial text representation.

Rather than assigning semantic labels independently, SignKG estimates the semantic type of each text segment under its spatial context. Let $c_i = (t_i, \rho_i, \mathcal{N}_\delta(s_i))$ denote the spatial context introduced in Section 3, where ρ_i is the normalized spatial descriptor and $\mathcal{N}_\delta(s_i)$ is the spatial neighborhood of segment s_i. The semantic type of s_i is then inferred as

$$\tau_i^* = \arg\max_{\tau \in \mathcal{T}} P(\tau|c_i). \tag{3}$$

Compared with conventional text classification, Eq. (3) explicitly incorporates spatial evidence into semantic inference. Consequently, two identical text fragments may be assigned different semantic types when their spatial contexts differ. This property is particularly important for Vietnamese signboards, where lexical ambiguity frequently occurs. For example, the phrase `HA NOI` may denote a business brand, a branch location, or a geographic address depending on its visual position and neighboring text segments. The proposed formulation naturally integrates heterogeneous information sources into a unified inference model. Let L_i, P_i, N_i, and R_i denote the lexical semantics, spatial position, neighborhood context, and business constraints associated with segment s_i, respectively. Assuming conditional independence among these evidence sources, the posterior probability can be approximated as

$$P(\tau_i|c_i) \propto P(L_i|\tau_i)P(P_i|\tau_i)P(N_i|\tau_i)P(R_i|\tau_i). \tag{4}$$

Equation (4) provides an intuitive interpretation of spatial-aware semantic inference. The lexical term evaluates whether the textual content is compatible with a particular entity type. The spatial term measures how likely the visual position supports that semantic interpretation. The neighborhood term captures contextual dependencies among nearby text segments. Finally, the rule term incorporates prior business knowledge, such as address patterns, phone number formats, category vocabularies, and service keywords. Combining these complementary evidence sources substantially improves semantic robustness under noisy OCR outputs. After estimating the semantic type, each text segment is converted into a typed business entity. Formally, the entity construction function is defined as $e_i = \Phi(t_i, \tau_i^*)$, where $\Phi(\cdot)$ transforms a recognized text segment into a structured business entity. The output of this stage is therefore the entity set $V(I) = \{e_1, e_2, \ldots, e_m\}$, which serves as the input for relation inference.

In the current implementation, lexical evidence is extracted using a lightweight large language model together with handcrafted business rules and regular expressions. Spatial evidence is derived from the normalized bounding boxes obtained during visual perception, while neighborhood relationships are computed from the spatial arrangement of text segments. Nevertheless, these implementation components merely instantiate the probabilistic formulation presented above. The core contribution of SignKG lies in modeling semantic extraction as a spatial-aware inference problem rather than as a standalone text classification task.

4.4 Spatial-aware Entity Linking

The objective of this stage is to establish semantic relations among the business entities extracted in the previous stage. Although semantic inference identifies the entity type of each text segment, the resulting entities remain isolated and do not yet form a coherent business representation. Entity linking therefore aims to determine which entities belong to the same business object and how they should be connected within the knowledge graph.

Let $V(I) = \{e_1, e_2, \ldots, e_m\}$ denote the entity set inferred from image I, where each entity is represented as $e_i = (v_i, \tau_i)$. The entity linking problem is formulated as estimating the optimal relation set $E^* = \arg\max_E P(E|V)$, where $E \subseteq V \times R \times V$ and R is the predefined relation vocabulary. Instead of predicting relations independently, SignKG constrains relation inference using both semantic compatibility and spatial consistency. Given two entities e_i and e_j, the optimal relation is obtained as $r_{ij}^* = \arg\max_{r \in R} P(r|e_i, e_j)$, subject to the compatibility constraint $\Gamma(\tau_i, \tau_j, r) = 1$, where $\Gamma(\cdot)$ verifies whether relation r is valid for the entity type pair (τ_i, τ_j). Consequently, invalid relations, such as linking a phone number to a category through $LOCATED_AT$, are automatically eliminated during inference.

To further improve structural consistency, SignKG incorporates spatial constraints inherited from the visual perception stage. Entities located within the same spatial neighborhood or sharing coherent layout patterns are assigned higher relation confidence than spatially unrelated entities. Therefore, relation inference jointly exploits semantic compatibility and spatial evidence rather than relying solely on textual information. The resulting relation set $E(I)$ provides the structural foundation for graph

construction by transforming isolated business entities into a semantically connected representation suitable for downstream knowledge graph generation.

4.5 Constraint-Guided Knowledge Graph Optimization

The previous stages estimate candidate business entities and semantic relations from a street signboard image. However, these predictions may still contain semantic ambiguity, OCR noise, duplicated entities, or incompatible relations. Directly assembling all predicted entities into a graph may therefore produce inconsistent business representations.

Instead of performing straightforward graph generation, SIGNKG formulates the final graph construction as a constrained optimization problem. The objective is to identify the most plausible urban business knowledge graph while simultaneously maximizing semantic confidence, preserving spatial consistency, and satisfying symbolic graph constraints.

Let $\hat{V} = \{e_1, e_2, \ldots, e_n\}$ denote the candidate business entities inferred in Section 4.3, and let $\hat{E} = \{(e_i, r, e_j)\}$ denote the candidate semantic relations obtained in Section 4.4. The objective is to estimate the optimal knowledge graph $G = (V, R, E)$ that maximizes the following graph objective $G^* = \arg\max_{G \in \mathcal{G}} J(G)$, where $J(G) = \alpha S_{\text{entity}} + \beta S_{\text{relation}} + \gamma S_{\text{spatial}} + \lambda S_{\text{constraint}}$, and $\alpha, \beta, \gamma, \lambda \geq 0$ are balancing coefficients. The first term evaluates the confidence of entity assignments $S_{\text{entity}} = \sum_{e_i \in V} \log P(\tau_i | c_i)$, where the posterior probability is estimated by the spatial-aware semantic inference model introduced in Section 4.3. The second term evaluates relation confidence $S_{\text{relation}} = \sum_{(e_i, r, e_j) \in E} \log P(r | e_i, e_j)$, which measures how well the inferred semantic relations explain the generated graph. To preserve the original layout information of the signboard, SIGNKG additionally introduces a spatial coherence score $S_{\text{spatial}} = -\sum_{(e_i, r, e_j) \in E} d(\rho_i, \rho_j)$, where $d(\rho_i, \rho_j)$ denotes the Euclidean distance between the normalized spatial descriptors of two related entities. This term encourages semantically related entities to remain spatially coherent.

Finally, graph consistency is modeled through symbolic constraints $S_{\text{constraint}} = -\eta_1 C_{\text{type}} - \eta_2 C_{\text{store}} - \eta_3 C_{\text{connect}}$, where $C_{\text{type}} = \sum_{(e_i, r, e_j) \in E} \big(1 - \Gamma(\tau_i, \tau_j, r)\big)$ penalizes incompatible semantic relations, $C_{\text{store}} = \max(0, N_{\text{store}} - 1)$ penalizes multiple dominant store entities extracted from a single signboard, and $C_{\text{connect}} = |\{e_i \in V : e_i \not\rightsquigarrow e_{\text{store}}\}|$ penalizes disconnected business entities that are not reachable from the corresponding store node. The optimization therefore seeks a graph that simultaneously: maximizes entity confidence, maximizes relation confidence, preserves spatial coherence, satisfies semantic compatibility, maintains a unique business anchor, and produces a structurally consistent business knowledge graph. Algorithm 1 summarizes the proposed graph optimization procedure.

Proposition 5. *Suppose that each iteration of Algorithm 1 exactly maximizes the entity assignment and relation assignment subproblems while enforcing the compatibility constraints through* Γ. *Then the graph objective satisfies* $J(G^{(t+1)}) \geq J(G^{(t)})$ *for every iteration* t.

Proof. During each iteration, the entity assignment step maximizes S_{entity} while fixing the remaining variables. Similarly, the relation assignment step maximizes S_{relation} un-

Algorithm 1 Constraint-Guided Knowledge Graph Optimization

Require: Candidate entity set $\widehat{V}$, candidate relation set $\widehat{E}$, relation vocabulary $\mathcal{R}$, compatibility function Γ, spatial descriptors $\{\rho_i\}_{i=1}^{n}$, convergence threshold ε
Ensure: Optimized urban business knowledge graph G^*

$V^{(0)} \leftarrow \widehat{V}$
$E^{(0)} \leftarrow \widehat{E}$
$G^{(0)} \leftarrow \big(V^{(0)}, \mathcal{R}, E^{(0)}\big)$
$J^{(0)} \leftarrow J\big(G^{(0)}\big)$
$t \leftarrow 0$
repeat
  $V^{(t+1)} \leftarrow \arg\max_{V \subseteq \widehat{V}} \sum_{e_i \in V} \log P(\tau_i \mid c_i)$
  $\widetilde{E}^{(t+1)} \leftarrow \arg\max_{E \subseteq \widehat{E}} \sum_{(e_i, r, e_j) \in E} \big[\log P(r \mid e_i, e_j) - \gamma d(\rho_i, \rho_j)\big]$
  $E^{(t+1)} \leftarrow \big\{(e_i, r, e_j) \in \widetilde{E}^{(t+1)} \;\big|\; \Gamma(\tau_i, \tau_j, r) = 1\big\}$
  $G^{(t+1)} \leftarrow \big(V^{(t+1)}, \mathcal{R}, E^{(t+1)}\big)$
  $J^{(t+1)} \leftarrow \alpha S_{\text{entity}}\big(G^{(t+1)}\big) + \beta S_{\text{relation}}\big(G^{(t+1)}\big) + \gamma S_{\text{spatial}}\big(G^{(t+1)}\big) - \lambda C\big(G^{(t+1)}\big)$
  $t \leftarrow t + 1$
until $\big|J^{(t)} - J^{(t-1)}\big| < \varepsilon$
$G^* \leftarrow G^{(t)}$
return G^*

der the compatibility constraints. The constraint filtering step never introduces invalid relations and therefore cannot decrease the constraint score. Consequently,$J(G^{(t+1)}) \geq J(G^{(t)})$. Hence the objective value is monotonically non-decreasing until convergence.

The proposed optimization framework unifies spatial reasoning, semantic inference, and symbolic constraints into a single objective function. Unlike conventional OCR pipelines that sequentially construct graph triples, SIGNKG performs constrained graph inference by jointly optimizing entity assignments, relation assignments, and graph consistency. The resulting graph therefore provides a mathematically grounded representation suitable for downstream urban knowledge applications.

4.6 Complexity Analysis

Let $n = |\widehat{V}|$ denote the number of candidate business entities extracted from a signboard and let $m = |\widehat{E}|$ denote the number of candidate semantic relations. The entity optimization step evaluates the posterior probability $P(\tau_i \mid c_i)$ for each candidate entity once during every iteration. Its computational complexity is therefore $T_{\text{entity}} = O(n)$. Similarly, the relation optimization step evaluates $P(r \mid e_i, e_j)$ for every candidate relation, leading to $T_{\text{relation}} = O(m)$. The compatibility filtering stage verifies the symbolic constraint $\Gamma(\tau_i, \tau_j, r)$ for each candidate relation exactly once, $T_\Gamma = O(m)$. Graph construction updates the entity set, relation set, and graph objective in linear time, $T_{\text{graph}} = O(n + m)$. Assuming that Algorithm 1 requires T optimization iterations before convergence, the total computational complexity is $T_{\text{total}} = O(T(n + m))$. Since

the optimization typically converges within a small number of iterations, the overall complexity is approximately linear with respect to the graph size. This makes SIGNKG suitable for large-scale urban business knowledge acquisition from street-level imagery.

5 Experimental Results

This section evaluates SIGNKG from five complementary perspectives: entity extraction, relation extraction, graph construction quality, and computational efficiency. The experiments are designed to answer the following research questions: ***RQ1:****How accurately does SIGNKG identify business entities from Vietnamese street signboards?*; ***RQ2:*** *How effectively does SIGNKG infer semantic relations among the extracted entities?* ; ***RQ3:*** *How much do spatial information and symbolic constraints contribute to the final performance?*; ***RQ4:*** *How consistent and computationally efficient are the generated knowledge graphs?;* ***RQ5:*** *How accurately and efficiently do the visual recognition and semantic extraction components transform signboard images into structured business records?*

5.1 Experimental Setup

Dataset The experimental dataset consists of Vietnamese street-level signboard images collected from diverse urban environments. Each image is annotated with spatial text segments, business entity types, semantic relations, and the corresponding knowledge graph triples. Let $\mathcal{D} = \{(I_k, S_k, V_k, E_k, G_k)\}_{k=1}^{N}$, where I_k denotes the input image, S_k is the set of spatial text segments, V_k is the annotated entity set, E_k is the annotated relation set, and G_k is the ground-truth business knowledge graph. The dataset is divided into training, validation, and testing subsets as $\mathcal{D} = \mathcal{D}_{\text{train}} \cup \mathcal{D}_{\text{val}} \cup \mathcal{D}_{\text{test}}$, with $\mathcal{D}_{\text{train}} \cap \mathcal{D}_{\text{val}} = \mathcal{D}_{\text{train}} \cap \mathcal{D}_{\text{test}} = \mathcal{D}_{\text{val}} \cap \mathcal{D}_{\text{test}} = \emptyset$.

Implementation Details

Implementation Details. The complete implementation consists of six sequential components. YOLOv8m with an input resolution of 640×640 is used to localize signboards, followed by perspective rectification. DBNet, implemented through PaddleOCR, detects individual text regions, and a domain-adapted VietOCR model recognizes Vietnamese text. A hybrid semantic extraction module combines spatial-aware regular expressions with a LoRA-adapted Qwen2.5-0.5B model to extract BRAND, SERVICE, ADDRESS, and PHONE fields. Finally, a TF–IDF-based Linear SVC assigns one of 18 business categories. The extracted fields, image coordinates, and geographic metadata are subsequently transformed into entities and semantic relations for constructing the urban business knowledge graph. The graph optimization parameters are denoted by $\Theta = \{\alpha, \beta, \gamma, \lambda, \eta_1, \eta_2, \eta_3, \delta, \varepsilon\}$, where δ is the spatial-neighborhood threshold and ε is the convergence tolerance.

Table 1: Summary of the Vietnamese street-signboard dataset used in the experiments.

Statistic	Value
Number of street-level images	302
Number of localized signboards	302
Number of annotated text regions	1,299
Number of business entities	298
Number of semantic relations	1,796
Number of knowledge graph triples	1,796
Number of entity types	7
Number of relation types	6
Number of business categories	18
Training/validation/test split	70/15/15%

End-to-End Processing Pipeline. The complete implementation consists of six sequential components. First, YOLOv8 is used to localize the signboard region from a street-level image. Second, a geometric rectification module corrects perspective distortion. Third, DBNet, implemented through PaddleOCR, detects individual text regions. Fourth, a fine-tuned VietOCR model recognizes Vietnamese text from the detected regions. Fifth, a hybrid semantic extraction module combines spatial-aware regular expressions with a LoRA-adapted Qwen2.5-0.5B model to extract the BRAND, SERVICE, ADDRESS, and PHONE fields. Finally, a TF–IDF-based Linear SVC assigns the business category. The extracted fields and geographic coordinates are subsequently transformed into entities and relations for constructing the urban business knowledge graph.

All images (see Table 1) were collected in real urban environments in Can Tho City, Vietnam. Each signboard was manually annotated with text bounding boxes, recognized text, entity labels, semantic relations, and graph triples. The annotations were reviewed by 05 annotators. Disagreements were resolved through discussion with a senior annotator.

5.2 Evaluation Metrics

Entity Extraction Metrics Let TP_e, FP_e, and FN_e denote the numbers of correctly predicted entities, incorrectly predicted entities, and missed entities, respectively. Entity precision, recall, and F_1-score are defined as $P_e = \frac{TP_e}{TP_e+FP_e}$, $R_e = \frac{TP_e}{TP_e+FN_e}$, and $F_1^e = \frac{2P_eR_e}{P_e+R_e}$.

Relation Extraction Metrics Let TP_r, FP_r, and FN_r denote the numbers of correctly predicted, incorrectly predicted, and missed relations. Relation precision, recall, and F_1-score are computed as $P_r = \frac{TP_r}{TP_r+FP_r}$, $R_r = \frac{TP_r}{TP_r+FN_r}$, and $F_1^r = \frac{2P_rR_r}{P_r+R_r}$.

Knowledge Graph Metrics A predicted triple is considered correct if its head entity, relation type, and tail entity all match the corresponding ground-truth triple. Let TP_g, FP_g, and FN_g denote the numbers of correct, incorrect, and missing graph triples.

Table 2: Comparison of YOLOv8 configurations for signboard localization. The best result is shown in bold.

Model	Input	Precision	Recall	mAP_{50}	$mAP_{50:95}$	Val. loss
YOLOv8n	640^2	0.85	0.77	0.87	0.70	0.71
YOLOv8n	1056^2	0.86	0.79	0.86	0.68	0.74
YOLOv8s	640^2	**0.87**	0.79	**0.88**	0.71	0.76
YOLOv8s	1056^2	0.81	**0.86**	0.87	0.69	0.76
YOLOv8m	640^2	0.86	0.82	0.87	**0.74**	**0.65**
YOLOv8m	1056^2	0.86	0.82	**0.88**	0.73	0.73

Triple precision, recall, and F_1-score are defined as $P_g = \frac{TP_g}{TP_g+FP_g}$, $R_g = \frac{TP_g}{TP_g+FN_g}$, and $F_1^g = \frac{2P_gR_g}{P_g+R_g}$. The valid-edge ratio is measured by $\text{VER} = \frac{\left|\left\{(e_i,r,e_j)\in E \mid \Gamma(\tau_i,\tau_j,r)=1\right\}\right|}{|E|}$. The store-connectivity ratio is defined as $\text{SCR} = \frac{|\{e_i \in V\setminus\{e_s\} \mid e_i \rightsquigarrow e_s\}|}{|V|-1}$, where e_s denotes the dominant store entity.

Efficiency Metrics The computational efficiency is evaluated using $T_{\text{avg}} = \frac{1}{N}\sum_{k=1}^{N} T_k$, where T_k is the processing time of the k-th image. The average number of optimization iterations is computed as $T_{\text{iter}} = \frac{1}{N}\sum_{k=1}^{N} t_k$.

5.3 Validation of Visual and Semantic Components

Before evaluating the generated knowledge graphs, we independently assess the principal upstream components. This analysis is important because errors in signboard localization, text detection, and text recognition propagate to entity and relation extraction.

Signboard Localization Table 2 compares six YOLOv8 configurations obtained by combining three model sizes with two input resolutions. YOLOv8m at 640×640 achieves the highest $mAP_{50:95}$ of 0.738 and the lowest validation box loss of 0.653. We therefore use this configuration in the complete SIGNKG pipeline.

As in Table 2, YOLOv8m at 640×640 achieves the highest $mAP_{50:95}$ of 0.738 and the lowest validation box loss of 0.653 among the evaluated YOLOv8 configurations. Although the 1056×1056 variant obtains a slightly higher mAP_{50}, its lower $mAP_{50:95}$ indicates less accurate localization under stricter intersection-over-union thresholds. We therefore select YOLOv8m at 640×640 for the complete pipeline.

Vietnamese Text Recognition We further evaluate the effect of domain-specific fine-tuning on VietOCR. The test set contains text-line images excluded from model training. Performance is measured using exact sequence accuracy and character error rate ($CER = \frac{S+D+I}{N_c}$,, where S, D, and I denote character substitutions, deletions, and insertions, respectively, and N_c is the number of characters in the ground-truth transcription).

As reported in Table 3, domain-specific fine-tuning increases exact sequence accuracy by 69.12 percentage points and reduces CER from 28.05% to 0.90%. This substantial improvement indicates that a generic Vietnamese OCR model is insufficient for

Table 3: Effect of domain-specific fine-tuning on Vietnamese signboard text recognition.

VietOCR variant	Sequence accuracy (%)	CER (%)
Pre-trained VietOCR	27.15	28.05
Fine-tuned VietOCR	**96.27**	**0.90**

Table 4: Field-level semantic extraction performance of the original and LoRA-adapted Qwen2.5-0.5B models.

Field	Base exact (%)	LoRA exact (%)	Base CER (%)	LoRA CER (%)
BRAND	4.0	**55.4**	82.9	**40.0**
SERVICE	5.0	**58.4**	109.4	**60.3**
OTHER	42.6	**52.5**	131.3	**115.4**

in-the-wild signboards containing decorative fonts, illumination changes, perspective distortion, and complex Vietnamese diacritics.

Structured Semantic Extraction The semantic extraction component is evaluated on 101 manually normalized signboards. Since phone numbers and addresses are extracted by deterministic spatial-aware rules, the language model is evaluated on the more ambiguous BRAND, SERVICE, and OTHER fields. We compare the original Qwen2.5-0.5B model with its LoRA-adapted counterpart.

The LoRA-adapted model consistently improves exact-match accuracy and reduces CER across all semantic fields. In addition, all outputs generated by the adapted model satisfy the required JSON syntax. The remaining exact-match errors are mainly caused by minor OCR errors, whitespace variation, and differences in entity-boundary annotation. For example, the model may extract "MM Mega Market" as a complete brand, whereas the reference annotation separates "MM" from "Mega Market". Such cases may remain semantically acceptable despite being penalized by exact matching.

Business Category Classification The extracted brand and service strings are concatenated, segmented using a Vietnamese word segmenter, and represented using TF–IDF features with one- to three-gram terms. Six conventional classifiers are compared for assigning each store to one of 18 business categories.

Linear SVC obtains the highest accuracy of 97.45% and is therefore adopted as the category-assignment component. The small difference among Linear SVC, Multinomial Naive Bayes, and Logistic Regression suggests that sparse lexical features provide strong evidence for business-category prediction.

5.4 Comparison with Baseline Methods

SIGNKG is compared with representative baseline approaches covering text-only extraction, rule-based extraction, layout-aware document understanding, and graph construction.

The comparison focuses not only on entity and relation accuracy, but also on the structural validity of the generated graphs.

Table 5: Performance of business-category classification methods on the independent test set.

Classifier	Accuracy (%)	Macro-F1 (%)
Decision Tree	88.08	88.13
K-Nearest Neighbors	90.19	89.87
Random Forest	92.79	92.73
Logistic Regression	97.29	97.53
Multinomial Naive Bayes	97.40	97.32
Linear SVC	**97.45**	**97.44**

Table 6: Comparison with baseline methods. The best results are shown in bold.

Method	P_e	R_e	F_1^e	F_1^r	F_1^g	**VER**	**SCR**
OCR + Regex	0.742	0.681	0.710	0.624	0.587	0.782	0.716
Text-only Classifier	0.801	0.774	0.787	0.706	0.674	0.824	0.781
Layout-aware Model	0.846	0.821	0.833	0.771	0.738	0.867	0.842
LLM-based Extraction	0.874	0.851	0.862	0.812	0.783	0.891	0.866
SIGNKG	**0.918**	**0.903**	**0.910**	**0.874**	**0.851**	**0.967**	**0.951**

5.5 Ablation Study

To quantify the contribution of each component, SIGNKG is evaluated after removing individual evidence sources and constraints.

Table 7: Ablation study of the proposed SIGNKG framework.

Variant	**Lexical**	**Spatial**	**Neighborhood**	**Rules**	F_1^e	F_1^g
SIGNKG	✓	✓	✓	✓	**0.910**	**0.851**
w/o Spatial	✓	×	✓	✓	0.864	0.794
w/o Neighborhood	✓	✓	×	✓	0.883	0.821
w/o Rules	✓	✓	✓	×	0.896	0.834
Text-only	✓	×	×	×	0.787	0.674

Table 7 evaluates the contribution of each component in SIGNKG. Removing the spatial module results in the largest performance degradation, reducing the entity-level F_1-score from 0.910 to 0.864 and the graph-level F_1-score from 0.851 to 0.794. This observation confirms that spatial layout provides the most informative evidence for resolving semantic ambiguity in Vietnamese street signboards. The neighborhood component also contributes noticeably to graph construction. Without neighborhood information, the graph-level F_1 decreases by 3.0 percentage points, indicating that nearby text segments provide valuable contextual cues for relation inference. Removing symbolic business rules causes a smaller yet consistent performance drop. Although lexical and

Table 8: Effect of spatial information on entity extraction.

Model	P_e	R_e	F_1^e
Text-only inference	0.801	0.774	0.787
Text + position	0.873	0.851	0.862
Text + position + neighborhood	**0.918**	**0.903**	**0.910**

spatial evidence remain available, compatibility constraints effectively eliminate invalid relations and improve graph consistency.

5.6 Effect of Spatial Information

The contribution of spatial context is evaluated by comparing the text-only formulation $\tau_i^* = \arg\max_{\tau \in \mathcal{T}} P(\tau \mid t_i)$ with the proposed spatial-aware formulation $\tau_i^* = \arg\max_{\tau \in \mathcal{T}} P(\tau \mid t_i, \rho_i, N_\delta(s_i))$.

5.7 Effect of Graph Constraints

The effect of symbolic constraints is evaluated by comparing graph construction with and without the compatibility function Γ and the store-connectivity constraint.

Table 9: Effect of symbolic graph constraints.

Configuration	F_1^r	F_1^g	**VER**	**SCR**
Without constraints	0.748	0.701	0.823	0.804
Type compatibility only	0.823	0.781	0.926	0.826
Type + connectivity	0.851	0.822	0.944	0.918
Full SIGNKG	**0.874**	**0.851**	**0.967**	**0.951**

Table 9 evaluates the contribution of symbolic graph constraints to relation inference and graph construction. Without any constraints, the generated graphs contain numerous incompatible edges, resulting in a graph-level F_1-score of only 0.701. Introducing the type compatibility function Γ substantially improves the valid-edge ratio from 0.823 to 0.926, confirming that symbolic filtering effectively removes invalid semantic relations. Adding the store-connectivity constraint further increases the store-connectivity ratio from 0.826 to 0.918 and improves the graph-level F_1-score to 0.822. Finally, the complete SIGNKG framework achieves the best overall performance, obtaining a relation-level F_1-score of 0.874, a graph-level F_1-score of 0.851, a valid-edge ratio of 0.967, and a store-connectivity ratio of 0.951. These results demonstrate that symbolic constraints and graph optimization significantly improve both semantic correctness and structural consistency of the generated knowledge graphs.

Table 10: Sensitivity to the spatial neighborhood threshold δ.

δ	F_1^e	F_1^r	F_1^g
0.10	0.892	0.843	0.818
0.20	0.904	0.862	0.838
0.30	**0.910**	**0.874**	**0.851**
0.40	0.907	0.868	0.844

Table 11: Average runtime of the complete image-to-graph processing pipeline on a CPU-based personal computer.

Component	Runtime (s)	Proportion (%)
YOLOv8 signboard localization	0.350	2.50
Perspective rectification	0.290	2.10
DBNet text-region detection	0.170	1.20
Fine-tuned VietOCR recognition	2.440	17.50
Qwen2.5-0.5B + spatial-aware rules	10.680	76.69
Linear SVC category classification	0.002	0.01
Complete pipeline	13.932	100.00

5.8 Sensitivity Analysis

The sensitivity of SIGNKG is analyzed with respect to the neighborhood threshold δ and the graph-objective coefficients. The effect of δ is evaluated over $\delta \in \{\delta_1, \delta_2, \ldots, \delta_q\}$. The graph-objective coefficients satisfy $\alpha + \beta + \gamma + \lambda = 1, \qquad \alpha, \beta, \gamma, \lambda \geq 0.$

Table 10 evaluates the influence of the spatial neighborhood threshold δ on entity extraction, relation inference, and graph construction. When δ is too small, only a limited number of neighboring text segments are considered, resulting in insufficient contextual evidence and lower extraction performance. Increasing δ consistently improves all evaluation metrics, with the best performance obtained at $\delta = 0.30$, where the entity-, relation-, and graph-level F_1-scores reach 0.910, 0.874, and 0.851, respectively. Further increasing the neighborhood threshold slightly decreases performance because irrelevant neighboring entities are incorporated, introducing semantic noise during spatial reasoning. These results indicate that an appropriate neighborhood size is essential for balancing contextual completeness and semantic discrimination. The graph consistency score may additionally be expressed as $Q(G) = \omega_1 \mathrm{VER} + \omega_2 \mathrm{SCR} + \omega_3 F_1^g$,

where $\omega_1 + \omega_2 + \omega_3 = 1$.

5.9 Computational Efficiency

We measure the runtime of the complete extraction pipeline on a CPU-based personal computer. Each value represents the average processing time per input image. Table 11 reports both the absolute runtime and the proportion contributed by each component.

The semantic extraction stage is the dominant computational bottleneck, accounting for approximately 76.7% of the total processing time. In contrast, graph-compatible category assignment using Linear SVC is negligible. The complete CPU-based pipeline

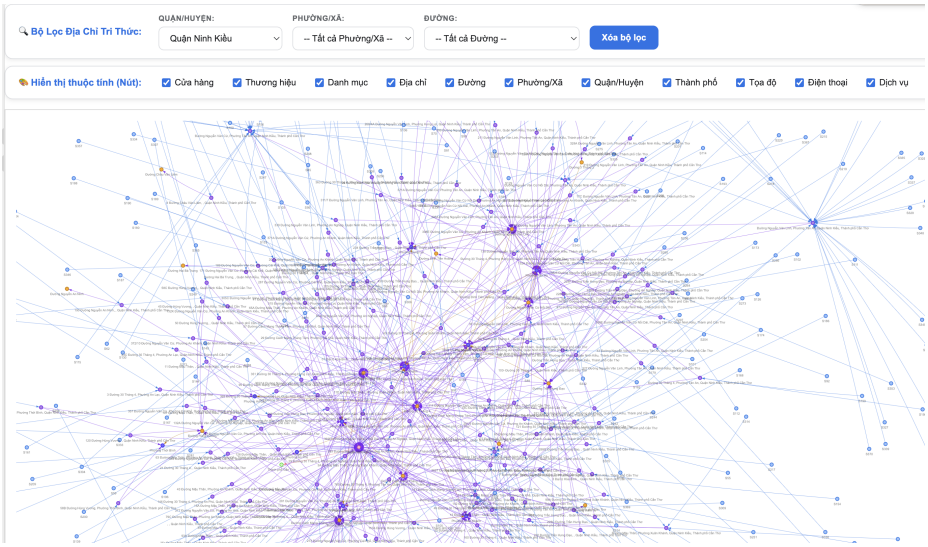


Fig. 2: Visualization of the aggregated urban business knowledge graph constructed from the extracted signboard information.

processes one image in approximately 13.93 seconds. This runtime is suitable for offline urban knowledge acquisition, although GPU acceleration, model quantization, and batched inference are required for real-time deployment.

5.10 Qualitative Results

Figure 2 visualizes the aggregated urban business knowledge graph constructed from the processed signboards. Nodes represent business entities and edges represent semantic relations defined by the SIGNKG.

5.11 Error Analysis

The end-to-end evaluation reveals three principal sources of error. First, DBNet may merge adjacent words, truncate decorative characters, or miss text located close to the signboard boundary. Second, residual OCR errors occur for heavily stylized fonts, three-dimensional characters, and low-contrast text. Third, semantic ambiguity remains when a phrase can represent either a brand, a service, or a location. These errors may propagate to entity assignment and subsequently affect graph relations.

A representative example is a signboard containing a long business name and several service descriptions. A minor text-region segmentation error may alter one or two OCR characters, causing the semantic extractor to omit the brand while still recovering the correct service and business category. This observation motivates jointly reporting component-level and graph-level performance rather than evaluating graph construction in isolation.

5.12 Discussion

The results show that SIGNKG benefits from the complementary use of lexical semantics, spatial layout, neighborhood context, and symbolic graph constraints. Spatial information provides the largest contribution to entity disambiguation, while type compatibility and store-connectivity constraints improve relation validity and graph consistency. The component-level experiments also clarify how upstream errors affect the final graph. Domain-specific VietOCR fine-tuning substantially improves text recognition, increasing exact sequence accuracy from 27.15% to 96.27%. However, high recognition accuracy on correctly detected text regions does not guarantee equivalent end-to-end performance because text detection failures may prevent relevant information from reaching the recognizer. The LoRA-adapted language model improves structured semantic extraction over the original compact model, but its exact-match results remain sensitive to OCR noise, whitespace variation, and entity-boundary differences. The hybrid design therefore assigns regular fields, such as phone numbers and addresses, to deterministic spatial-aware rules, while the language model focuses on context-dependent fields such as brands and services. The current CPU implementation requires approximately 13.93 seconds per image, with semantic extraction accounting for most of the runtime. Consequently, the present system is more appropriate for offline or incremental urban knowledge acquisition than for real-time deployment. Model quantization, constrained decoding, distillation, and GPU inference are promising directions for reducing computational cost.

6 Conclusion

This paper presented SIGNKG, a spatial-aware framework for constructing urban business knowledge graphs from Vietnamese street signboards. Unlike conventional OCR pipelines that produce isolated text strings, SIGNKG integrates visual perception, spatial-aware semantic inference, constrained entity linking, and graph optimization to transform street-level images into structured business knowledge.

The experimental results indicate that spatial layout and neighborhood context substantially improve entity disambiguation, while symbolic compatibility and connectivity constraints enhance graph validity and structural consistency. Component-level evaluation further demonstrates the importance of domain-specific OCR adaptation and hybrid semantic extraction for Vietnamese in-the-wild signboards.

Future work will focus on expanding the annotated dataset, improving text-region detection for decorative and irregular layouts, reducing the computational cost of semantic extraction, handling signboards containing multiple businesses, and evaluating SIGNKG in larger urban mapping and business-search applications.

References